

Resolution-Aware Lexicographic Tail-Regret Selection under Uncertain Normalisation Bounds

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Abstract

Selecting a recommendation from a multicriteria candidate set requires a declared preference rule and a treatment of uncertain normalisation bounds. Exact lexicographic minimax gives arbitrarily small worst-case differences absolute priority, hard to justify at finite decision resolution. We introduce ReLexTail, a preorder that first compares fixed-resolution categories of maximum and upper-tail active-range probe disappointments, then applies exact tail and ordered-profile refinements. The construction is complete and transitive, Pareto-compatible under monotone separating probes, and invariant to uniform replication of the probe multiset. A sufficient category-stability condition is distinguished from stability of the fully refined recommendation, and a sequential linear and mixed-integer characterisation covers continuous polyhedra. We then certify the decision rather than every coordinate. We prove that a recommendation can be certified throughout a bound box with no candidate retaining any category, and that keeping a probe’s shared normalisation anchors inside a single quotient yields an enclosure contained in the independent one. Rational examples show both gaps are real: the all-candidate condition certifies nothing on an instance whose decision is certain over the whole box, and anchor slack alone decides whether another box is certified. On a benchmark of 192 candidate sets with splits locked before the run, the two changes raise the certified decision radius by 0.0330 and 0.0204 at a matched budget, and an exact oracle and an independent replay verifier falsified none of the emitted certificates. A separate 90-instance study with 500 matched perturbations at eight levels and 10,000 held-out preference vectors gives 19.30% point changes at resolution 0.02 against 20.88% for exact probe-lexicographic selection, with a paired interval including zero, and mean upper-tail regret 0.5411 against 0.5428. Grid shifts, ablations and a public building-simulation check expose boundary dependence. Decision quality against external preference populations and any calibration guarantee remain untested.

Keywords: multicriteria decision analysis; lexicographic selection; regret; conditional value-at-risk; normalisation uncertainty; robustness

1 Introduction

A multicriteria analysis often produces several efficient alternatives while a decision maker ultimately needs a recommendation. Selecting one alternative requires assumptions about acceptable trade-offs, even when those assumptions are expressed without an explicit criterion-weight vector. Weighted sums, reference-point methods and distance-based methods provide different ways of making these assumptions operational (Hwang & Yoon, 1981; Miettinen, 1999; Wierzbicki, 1980). Their recommendations also depend on the numerical scales on which criteria are compared. When normalisation bounds are estimated, robustness to those bounds is a distinct concern from uncertainty about preferences.

This paper considers a declared family of monotone questions about each alternative, called probes. A probe can assess one criterion, the mean across criteria, or the worst criterion. Its value is transformed into an active-range disappointment: zero denotes the best probe value available in the fixed candidate set and one the worst. A decision maker can inspect the named concerns

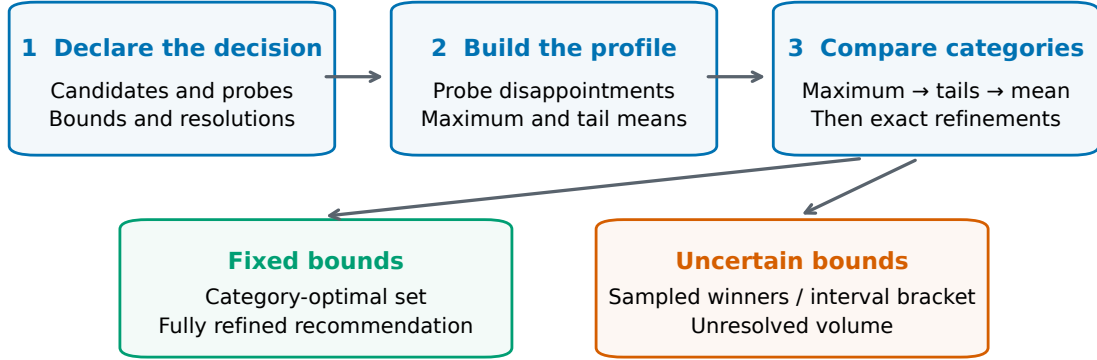


Figure 1: Overview of the decision procedure and graphical abstract. The declared model determines active-range disappointments. Category comparisons precede exact refinement. Under uncertain bounds, sampling and interval certification answer different questions and require different reports.

before accepting the recommendation. The probe family, its multiplicities, the candidate set and the normalisation convention remain substantive modelling choices.

Exact lexicographic minimax orders the sorted disappointments from worst to best. It gives any improvement in the largest disappointment priority over all improvements below it, regardless of the size of that leading difference. This is appropriate when priorities are strict. It can be difficult to justify when the decision model has a finite reporting resolution. Pairwise difference tolerances are an imperfect remedy because approximate equality need not be transitive. We instead use fixed resolution categories, followed by exact refinements, so that the comparison is a lexicographic order on a single well-defined vector.

The resulting rule, ReLexTail, first compares categories of the maximum and of upper-tail mean disappointments. Only when all category coordinates agree does it compare exact tail means and then the full sorted profile. Its contribution is this explicit hierarchy together with its decision-aiding interpretation, finite-set properties and treatment of uncertain normalisation bounds. The building blocks, including ordered aggregation, empirical conditional value-at-risk (CVaR) and lexicographic optimisation, are established tools. We do not claim their invention or a general superiority over other MCDA approaches.

Ordered weighted averaging assigns weights to ordered outcomes rather than to named criteria (Yager, 1988). Linear formulations of ordered aggregation and sums of the largest functions provide useful computational machinery (Ogryczak & Śliwiński, 2003; Ogryczak & Tamir, 2003). Equitable optimisation studies related order-based preferences (Kostreva & Ogryczak, 1999). ReLexTail uses ordered quantities to evaluate a declared multiset of concerns. Its categories postpone sub-resolution differences, but its exact refinements still distinguish candidates within a category-optimal set. Accordingly, the categories do not create permanent indifference in the final preorder.

CVaR provides a continuous summary of an upper tail, including fractional empirical atoms (Rockafellar & Uryasev, 2000). Here its atoms are probe disappointments, not uncertain future outcomes. A separate CVaR is used only in experimental evaluation, across held-out preference vectors. Keeping these two distributions distinct is essential: a small tail of declared concerns need not imply a small tail of regret for an external decision maker.

Stochastic multicriteria acceptability analysis explores the support for alternatives under uncertain preference parameters and data (Lahdelma, Hokkanen, & Salminen, 1998; Tervonen & Figueira, 2008). Robust ordinal regression considers compatible value functions and distinguishes necessary from possible preferences (Greco, Mousseau, & Śłowiński, 2008). Our possible and necessary winner sets use similar quantifiers but vary normalisation bounds while holding the declared preference rule fixed. They therefore complement preference-uncertainty analysis rather than replace it. Likewise, exact LexPR in this paper denotes a reference order on the same disappointment representation. It is a baseline definition, not a claim that lexicographic minimax itself is new.

Recent work reinforces four adjacent needs without making them interchangeable. Robustness coefficients diagnose ranking sensitivity (Paradowski, Wątróbski, & Sałabun, 2025); unknown-weight

Table 1: Structural positioning of neighbouring approaches. Entries describe the primary role of each family, not a quality ranking; implementations within a family can add further capabilities.

Approach	Primary modelling object	Primary uncertainty	Typical output
Weighted score/distance	Compensatory score or distance	Weights/data if varied	Score or ranking
Leximax / LexPR	Ordered worst-to-best profile	None in the order	Exact optimal class
SMAA	Distribution over preferences/data	Preference/data sampling	Acceptability indices
Robust ordinal regression	Compatible preference models	Preference imprecision	Necessary/possible relations
DRO	Worst distribution in an ambiguity set	Probability-law ambiguity	Robust decision
ReLexTail	Fixed cells, tail summaries, exact profile	Bounded normalisation parameters	Category/exact winner sets

procedures explore preference uncertainty (Więckowski & Sałabun, 2025); Bayesian pairwise queries support interactive elicitation (Huber, Rojas Gonzalez, & Astudillo, 2025); and contrastive records can explain why one alternative is preferred to another (Erwig & Kumar, 2025). Distributionally robust optimisation instead protects against ambiguity in a probability law (Kuhn, Shafiee, & Wiesemann, 2025). ReLexTail addresses none of these problems in full generality: its primary uncertainty is a bounded perturbation of normalisation parameters, and its explanation is a deterministic replay of the declared profile.

Certifying a recommendation under uncertain bounds raises a question that is easy to answer too conservatively. The natural sufficient condition asks every candidate to keep every risk coordinate inside its resolution cell, and it fails whenever any coordinate touches a cell boundary – including a coordinate belonging to an alternative that is not in contention, a coordinate that is structurally invariant, and a coordinate occurring after the comparison has already been decided. None of those motions can change the recommendation. A second, independent source of conservatism sits in the arithmetic: the two normalisation anchors of a probe are shared by all candidates at a given bound vector, and widening the numerator and denominator of the disappointment ratio into separate intervals lets each take an extreme value the shared anchor could not have produced. Section 5 replaces both, proves that neither replacement weakens soundness, and measures what each is worth on a prespecified benchmark with locked splits.

The contribution is eightfold. First, we specify a fixed-cell tail-regret preorder with exact refinement, including boundary and tie conventions. Second, we establish finite-set existence, transitivity, conditional Pareto compatibility, replication invariance and a category-stability condition. Third, we distinguish sampled sensitivity from sound interval possible-winner elimination. Fourth, we give a sequential polyhedral formulation and its stage count. Fifth, we prove a decision-focused certification condition that requires no category to be retained, and exhibit a rational instance on which the all-candidate condition certifies nothing while the decision-focused condition certifies the entire uncertainty box without subdivision. Sixth, we prove that keeping the shared probe anchors inside a single quotient yields an enclosure contained in – and generically strictly inside – the independent one, and exhibit a rational instance where that difference decides whether the box is certified at all. Seventh, we test the quality–stability trade-off with matched perturbations, held-out preferences, paired uncertainty summaries, grid-origin sensitivity and ablations on 90 synthetic sets, plus a public building-simulation demonstration, and we test the certification claims on a separate locked benchmark with an exact rational oracle and an independent replay verifier. Eighth, we produce contrastive decision records and a complete replay package. These contributions combine established ordered-risk tools around finite decision resolution; they do not imply universal superiority or operational validation. The certification results say nothing about decision quality, external regret or calibration, which are not tested here. A ten-supplier illustration supplies an interpretable worked decision.

2 Representation and resolution-aware selection

2.1 Declared probes and active-range disappointment

Let $A = \{x_1, \dots, x_N\}$ be a fixed, nonempty finite candidate set, and let $f_i(x)$, $i = 1, \dots, m$, be criteria to minimise. For bounds $b = (z^*, z^{\text{nad}})$ with positive denominators, define

$$r_i(x; b) = \frac{f_i(x) - z_i^*}{z_i^{\text{nad}} - z_i^*}. \quad (1)$$

Nominal bounds are the minimum and maximum of each criterion over A . Thus nominal values lie in $[0, 1]$. Perturbed or external bounds need not enclose the observed criteria, so perturbed values may lie outside that interval. We do not clip them. Constant criteria are excluded before normalisation and their exclusion is reported.

A probe q is finite-valued and coordinatewise nondecreasing on a domain containing every reachable normalised vector. The declared family $\mathcal{Q}$ is a multiset. Its multiplicities act as weights in the empirical distribution of probe values. For each bound vector, define

$$q^*(b) = \min_{x \in A} q(r(x; b)), \quad q^w(b) = \max_{x \in A} q(r(x; b)), \quad R_q(b) = q^w(b) - q^*(b).$$

Retain probes with $R_q(b) > \varepsilon_q$, and set

$$D_q(x; b) = \frac{q(r(x; b)) - q^*(b)}{R_q(b)} \in [0, 1]. \quad (2)$$

The nominal computation uses $\varepsilon_q = 10^{-9}$. The ratio is evaluated without adding this threshold to the denominator. The threshold determines retention; it is not a smoothing constant. A run with no retained probes returns the entire candidate set as equivalent and cannot support a discriminating recommendation. All results below concern a nonempty retained family. Certification additionally requires a fixed retained family over its uncertainty box.

The canonical family contains single-criterion probes $q_i(r) = r_i$, the mean $q_\mu(r) = m^{-1} \sum_i r_i$, and the maximum $q_{\max}(r) = \max_i r_i$. There are at most $m + 2$ retained probes. The mean may be constant on a particular candidate set and then contributes no information. We report the retained count rather than treating $m + 2$ as an unconditional identity. Correlated criteria are not automatically merged, and no cluster probes are used in the computational study.

For a retained singleton, the two normalisations cancel exactly:

$$D_i(x; b) = \frac{f_i(x) - \min_{y \in A} f_i(y)}{\max_{y \in A} f_i(y) - \min_{y \in A} f_i(y)}. \quad (3)$$

Consequently, singleton disappointments are invariant to admissible bounds. Aggregate disappointments generally are not. Fixing external criterion bounds alone also does not provide independence from edits to A , because the probe anchors in Equation (2) still depend on that set. Full candidate-set independence would require fixing those anchors as well.

2.2 Tail means, categories and exact refinement

Write $D_{[1]}(x) \geq \dots \geq D_{[K]}(x)$ for the retained disappointments sorted in decreasing order and $M(x) = D_{[1]}(x)$. For a tail fraction $\alpha \in (0, 1]$, define

$$T_\alpha(x) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{\alpha K} \sum_{q=1}^K [D_q(x) - t]_+ \right\}. \quad (4)$$

Here α denotes the mass of the upper tail, rather than its confidence level. For $h = \alpha K$, $k = \lfloor h \rfloor$ and $\theta = h - k$,

$$T_\alpha(x) = \frac{\sum_{j=1}^k D_{[j]}(x) + \theta D_{[k+1]}(x)}{h}.$$

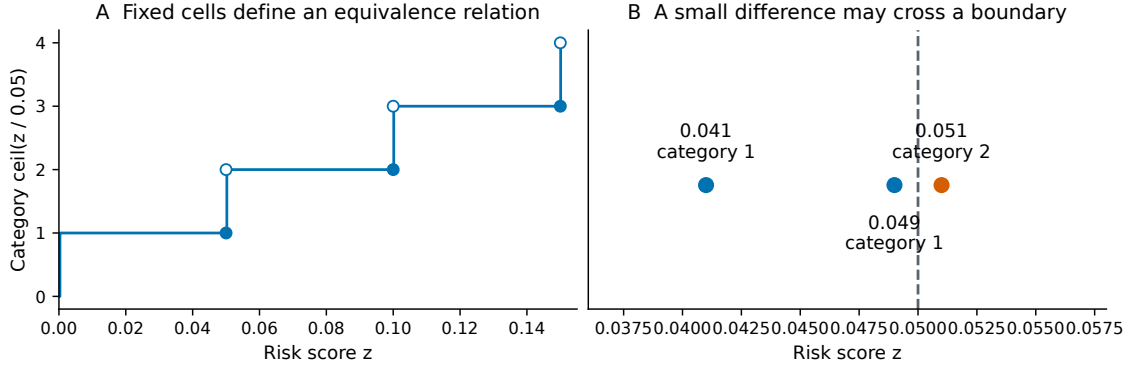


Figure 2: Fixed resolution cells and boundary sensitivity, illustrated at $\delta = 0.05$. Filled endpoints belong to the lower category and open endpoints to the upper category. Scores 0.049 and 0.051 are close but occupy different cells. This is a constructed explanation of the rule, not an empirical result.

The fractional term is omitted when $\theta = 0$. If $h < 1$, the expression equals $M(x)$. The fraction $\alpha = 1$ gives the overall mean. We use $T_{25} = T_{0.25}$, $T_{50} = T_{0.50}$ and $T_{100} = T_1$. Small retained families may make some tail coordinates redundant, without invalidating the rule.

Declare positive widths $\delta_M, \delta_{25}, \delta_{50}, \delta_{100}$. Categories are

$$B_\delta(z) = \lceil z/\delta \rceil, \quad C_\ell(x) = B_{\delta_\ell}(R_\ell(x)), \quad R = (M, T_{25}, T_{50}, T_{100}). \quad (5)$$

Category zero contains zero. For $j \geq 1$, category j is the interval $((j-1)\delta, j\delta]$. The boundary convention is part of the model and must be preserved in numerical replay.

For numerical replay, the implementation evaluates the quotient once in binary double precision, snaps it to the nearest integer only when the discrepancy is at most eight machine epsilons times $\max(1, |z/\delta|)$, and then applies the ceiling. This rule protects declared decimal boundaries such as 0.10/0.05 from representation noise; it is not a decision tolerance and does not move observations that are substantively near a boundary. Stable row identifiers settle any remaining full-profile tie.

2.3 Eliciting and auditing resolution widths

A resolution width is best interpreted as the smallest difference on an active-range concern that the decision maker is willing to give categorical priority. It is not estimated from the alternative that happens to win. One practical protocol is: (i) name the probes and show their best and worst attainable physical values; (ii) present controlled pairs that differ on one risk summary while other summaries are held as nearly equal as feasible; (iii) bracket the smallest difference that warrants crossing a priority level; and (iv) translate that bracket to the $[0, 1]$ active-range scale. Distinct widths may be elicited for maximum, upper-quarter, upper-half and mean concern. If the elicited answer is an interval, both endpoints and nearby values should be analysed rather than replaced by a favourable point estimate.

The reported analysis must then vary widths and the placement of cell boundaries. For a diagnostic origin fraction $\gamma \in [0, 1)$, we use

$$B_{\delta, \gamma}(z) = \left\lceil \frac{z - \gamma\delta}{\delta} \right\rceil,$$

with the same closed-upper-endpoint convention; because $z \geq 0$ and $\gamma < 1$, the first category remains labelled zero. The main specification is $\gamma = 0$. Shifts $\gamma \in \{0, .25, .5, .75\}$ test whether conclusions depend on an accidental boundary placement. This diagnostic does not substitute for eliciting a defensible physical resolution.

Define the full profile

$$\Psi(x) = (C_M, C_{25}, C_{50}, C_{100}, T_{25}, T_{50}, T_{100}, M, D_{[2]}, \dots, D_{[K]})(x), \quad (6)$$

$$x \preceq_{\text{ReLexTail}} y \iff \Psi(x) \leq_{\text{lex}} \Psi(y).$$

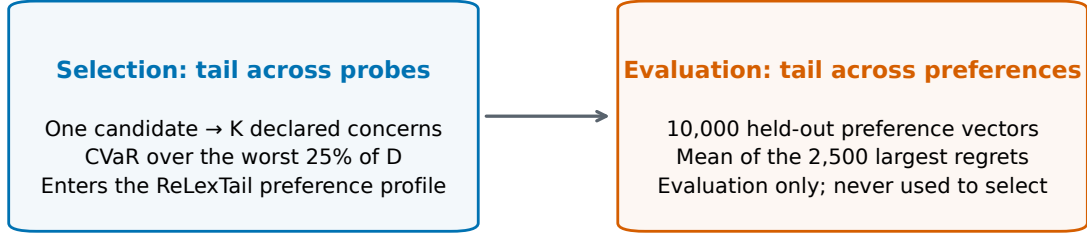


Figure 3: Two distinct uses of an upper tail. Selection aggregates the declared probes of a candidate. Evaluation aggregates range-normalised regret across held-out preference vectors. The latter weights are shared across methods and are never used to select a candidate.

Every category coordinate precedes every exact coordinate. Placing exact M immediately after C_M would restore strict maximum priority within each maximum category, defeating the intended comparison of broader tails. Exact refinement is instead postponed until the entire category prefix has been compared.

Let W_{cat} minimise the four-coordinate category prefix and W_{exact} minimise the full profile. Then $W_{\text{exact}} \subseteq W_{\text{cat}}$. Distinct alternatives are equivalent under the full preorder exactly when their sorted disappointment profiles coincide. When a single point is needed, we select the smallest candidate identifier in W_{exact} and report this convention separately from the preference rule. The exact LexPR baseline minimises $D^\downarrow$ directly; deleting the category coordinates alone from Equation (6) does not produce LexPR, because the exact tail coordinates would still precede M .

Fixed cells resolve a logical issue, not all sensitivity. For scalar minimisation with $\delta = 0.05$, values 0, 0.04 and 0.08 give adjacent pairs within tolerance, whereas the first value is strictly preferable to the third. This demonstrates nontransitive approximate indifference, without establishing a strict preference cycle. Category equality is transitive. However, a very small change across a fixed boundary can alter a category, and exact refinement can change the point while the category-optimal set stays fixed. We therefore measure these outcomes separately.

3 Properties and interpretation

Theorem 1 (Existence and completeness). *For every finite nonempty candidate set A , fixed finite retained probe family, and positive declared resolution vector, a ReLexTail-optimal alternative exists, and $\preceq_{\text{ReLexTail}}$ is a complete preorder.*

Proof. $\Psi(x)$ is a finite-dimensional vector. Lexicographic order on these vectors is a complete preorder (not a total order, as distinct alternatives can have identical Ψ). The pullback through Ψ is a complete preorder, which has at least one minimal element on a finite non-empty set. $\square$

Theorem 2 (Positional anonymity conditional on the declared multiset). *For every permutation σ of the probe coordinates under fixed probe multiplicities, $\Psi(\sigma D) = \Psi(D)$.*

Proof. M , empirical CVaR, and resolution categories depend only on symmetric quantities (order statistics). $D^\downarrow$ is permutation-invariant. Selective duplication changes the multiset and can change the result. $\square$

Proposition 1 (Strict profile monotonicity). *If $D(x) \leq D(y)$, then $x \preceq_{\text{ReLexTail}} y$. If strictly less somewhere, $x \prec_{\text{ReLexTail}} y$.*

Proof. Every coordinate of Ψ is a nondecreasing functional of the coordinatewise input: empirical CVaR is monotone, B_δ is nondecreasing, and sorting is monotone, so $D(x) \leq D(y)$ implies $\Psi(x) \leq \Psi(y)$ coordinatewise, hence $\Psi(x) \leq_{\text{lex}} \Psi(y)$. If additionally $D_q(x) < D_q(y)$ for some

q , then $T_{100}(x) < T_{100}(y)$, and no coordinate preceding T_{100} in Ψ can strictly favour y , so $\Psi(x) <_{\text{lex}} \Psi(y)$. $\square$

Theorem 3 (Criterion-level Pareto compatibility). *Fix a retained probe family Q . Assume every $q \in Q$ is coordinatewise nondecreasing and Q is Pareto-separating on the reachable domain: whenever $r(x) \leq r(y)$ and $r(x) \neq r(y)$, there exists $q \in Q$ such that $q(r(x)) < q(r(y))$. Then criterion-level Pareto dominance of x over y implies $x \prec_{\text{ReLexTail}} y$.*

Proof. Since x Pareto-dominates y and the retained probes are nondecreasing, we have $D_q(x) \leq D_q(y)$ for all q . By monotonicity of empirical CVaR, $T_\alpha(x) \leq T_\alpha(y)$ for all α , and since B_δ is nondecreasing, $C_\alpha(x) \leq C_\alpha(y)$. Thus $\Psi(x) \leq_{\text{lex}} \Psi(y)$ and weak Pareto compatibility holds. For strictness, the Pareto-separating assumption gives a retained probe q with $q(r(x)) < q(r(y))$. On that probe the active-range map $q(r(x)) \mapsto D_q(x)$ is a strictly increasing affine map (its slope is $1/R_q > 0$ because the probe is retained), so $D_q(x) < D_q(y)$, hence $D(x) \neq D(y)$. Since $D(x) \leq D(y)$ coordinatewise with at least one strict coordinate, $T_{100}(x) = \frac{1}{K} \sum_q D_q(x) < T_{100}(y)$. The coordinate T_{100} precedes the terminal sorted block in Ψ and no earlier coordinate can favour y , so $\Psi(x) <_{\text{lex}} \Psi(y)$ and $x \prec_{\text{ReLexTail}} y$. $\square$

Theorem 4 (Uniform replication invariance). *Let rD denote the profile obtained by repeating every disappointment r times. Under identical resolutions δ and tail fractions α , $D \preceq_{\text{ReLexTail}} E \iff rD \preceq_{\text{ReLexTail}} rE$.*

Proof. The proof distinguishes the empirical distribution block from the terminal sorted-profile block. First, the empirical distribution of rD is identical to D , meaning the empirical functionals remain unchanged: $M(rD) = M(D)$ and $T_\alpha(rD) = T_\alpha(D)$, so all resolution categories coincide. Second, if $D^\downarrow(x) <_{\text{lex}} E^\downarrow(y)$, the uniformly repeated sorted vectors preserve the first differing original coordinate and its direction, preserving the lexicographic comparison. $\square$

Theorem 5 (Category stability). *For any finite candidate set A and fixed retained probe count K , empirical upper-tail CVaR and the maximum are 1-Lipschitz in the supremum norm over disappointment space. For any risk coordinate $R_\ell \in \{M, T_{25}, T_{50}, T_{100}\}$, define the distance to the nearest category boundary as $\mu_\ell(x) = \text{dist}(R_\ell(x), \delta_\ell \mathbb{Z})$. Let $\mu_{\text{cell}} = \min_{x \in A, \ell} \mu_\ell(x)$. If a perturbation in the disappointment matrix satisfies $\|\hat{D} - D\|_\infty = \eta < \mu_{\text{cell}}$, then every candidate remains in the same category on every category coordinate. Note that if perturbations occur in the bounds b rather than directly in disappointments, a propagation bound $\|D(b) - D(\hat{b})\|_\infty \leq L_B \|b - \hat{b}\|$ must be applied.*

Proof. For the maximum M , the 1-Lipschitz property is immediate since $|M(D) - M(\hat{D})| \leq \max_q |D_q - \hat{D}_q| \leq \eta$. For empirical upper-tail CVaR, by definition of the perturbation bound, $D - \eta \mathbf{1} \leq \hat{D} \leq D + \eta \mathbf{1}$. By monotonicity and translation equivariance of CVaR, $T_\alpha(D) - \eta \leq T_\alpha(\hat{D}) \leq T_\alpha(D) + \eta$, yielding $|T_\alpha(D) - T_\alpha(\hat{D})| \leq \eta$. Because the shift in every R_ℓ is at most $\eta < \mu_{\text{cell}}$, no value crosses a category boundary $\delta_\ell \mathbb{Z}$, preserving all categories. $\square$

Remark (Exact-refinement fragility). Stability of the fully refined ReLexTail point does not generally follow from a positive gap at the nominal decisive exact coordinate. If any preceding exact coordinate is tied and can change under perturbation, an arbitrarily small perturbation may reverse the lexicographic comparison. A positive full-winner radius requires additional structural invariance of the preceding ties or a direct interval-profile certificate.

The category condition is sufficient, not necessary. It can be vacuous when any risk coordinate lies exactly on a boundary, even if that coordinate is structurally invariant. For perturbations in physical bounds, a bound on the induced change in D is additionally required. A distance measured only for one coordinate of the selected candidate cannot substitute for the minimum over all candidates and risk coordinates.

Nor are category-optimal sets generally nested when a common width is increased: cell boundaries move as well as merge, and later category coordinates can become decisive in a different way. Nesting follows only for specially aligned coarsenings at the level of each scalar partition, and even then it does not automatically transfer to the lexicographically optimal set. Resolution sensitivity must therefore be computed rather than inferred from set inclusion.

Table 2: Uncertainty and tolerance sources in the reported workflow.

Quantity	Role	Treatment
Resolution δ	Preference-model granularity	Declared fixed cells; varied in sensitivity analysis
Bound level p	Epistemic normalisation uncertainty	Matched random draws or complete interval box
Retention ε_q	Degenerate-probe guard	Fixed at 10^{-9} ; never added to a denominator
Machine/solver tolerance	Numerical implementation	Boundary snapping by eight machine epsilons for finite replay; solver tolerances must be reported separately

These properties describe the declared preorder and do not imply preference recovery, independence of the candidate set, or statistical optimality under an external loss distribution. In particular, equal replication of every probe preserves the empirical distribution, whereas duplicating only a subset does not.

4 Computation and uncertain bounds

4.1 Finite selection and comparison records

For finite A , evaluate each retained probe, compute its anchors, form D , sort each row, construct Ψ and scan for the minimum. A second scan collects every exact and category tie. Computing the canonical family costs $O(Nm)$ and sorting costs $O(NK \log K)$. Comparing profiles in a scan costs $O(NK)$. Thus selection costs $O(Nm \log m)$ for the canonical family. This complexity excludes generating candidates and Pareto filtering. It is an operation count, not a measured runtime guarantee.

Four numerical quantities have different meanings and are never substituted for one another. Table 2 records the distinction used in the algorithms and experiments.

Finite replay uses IEEE binary double precision for raw arithmetic, descending stable sorting, the declared boundary snapping rule in Section 2, and the smallest immutable row identifier only after equality of the full profile. The independent certification checks recompute category profiles from exact rational representations of outward-rounded interval endpoints. These conventions are recorded in the protocol file so that numerical tolerance cannot silently become a preference parameter.

For each rival y of a unique winner x^* , a contrastive record stores the first differing coordinate of $\Psi(x^*)$ and $\Psi(y)$, the two values, the comparison direction and the resolution if that coordinate is categorical. Together with the full disappointment matrix, this permits independent comparison replay. A labelled disappointment plot alone explains concerns but does not prove optimality. Recomputing probe values additionally requires the criterion matrix, bounds, formulas and retained identifiers. Recreating the candidates requires the upstream model and seeds. These levels of replay should not be conflated.

4.2 Sampling and certified enclosures

For a compact admissible box $\mathcal{B}$, let $W(b)$ be the full exact optimal class. Define

$$S^{\text{pos}} = \bigcup_{b \in \mathcal{B}} W(b), \quad S^{\text{nec}} = \bigcap_{b \in \mathcal{B}} W(b). \quad (7)$$

A sampled union is an inner approximation of the possible set. A sampled intersection is an outer approximation of the necessary set, not a certificate of necessary optimality. If a candidate wins on a region of sampling probability at least $a > 0$, then B independent draws miss it with probability at most $(1 - a)^B$. By a union bound, $B \geq \lceil \log(N/\epsilon) / [-\log(1 - a)] \rceil$ detects all such candidates with probability at least $1 - \epsilon$. It gives no guarantee for regions of zero or smaller probability.

Our experiments perturb each bound by a fraction of its nominal criterion range. If $I_i = \min_A f_i$, $N_i = \max_A f_i$ and $s_i = N_i - I_i > 0$, define

$$\mathcal{B}(p) = \prod_i [I_i - ps_i, I_i + ps_i] \times [N_i - ps_i, N_i + ps_i], \quad 0 \leq p < \frac{1}{2}. \quad (8)$$

Denominators are bounded below by $(1 - 2p)s_i > 0$. This convention is invariant to a change of criterion origin. Frequency estimates use independent uniform draws for each bound coordinate. Certification concerns the complete box and does not depend on this sampling distribution. The supplier sampling illustration also includes $p = 0.5$, for which sampled denominators were checked positive; that endpoint is not used for certification because the closed box can contain zero denominators.

Consider a sub-box β . Assume every retained probe has active range uniformly greater than ε_q throughout β . Singleton retention can be checked using

$$\inf_{b \in \beta} R_{q_i}(b) = \frac{s_i}{\sup_{b \in \beta} (z_i^{\text{nad}} - z_i^*)}.$$

A nominal check alone is insufficient. For aggregates, bound $q(r(x; b))$ using coordinatewise enclosures and monotonicity, then enclose the anchors and the ratio in Equation (2). The extrema of each normalised criterion over its two-dimensional rectangle occur at corners when denominators remain positive. Dependency between the ratio terms may produce substantial slack. If uniform retention cannot be established, the box remains unresolved and is subdivided or returned with its inherited candidates.

Proposition 2 (Sound elimination). *Suppose the retained multiset is fixed and $D^L(x; \beta) \leq D(x; b) \leq D^U(x; \beta)$ for all $b \in \beta$. If*

$$\Psi(D^U(x; \beta)) <_{\text{lex}} \Psi(D^L(y; \beta)),$$

then $x \prec_{\text{ReLexTail}} y$ throughout β .

Proof. Maximum, tail means, category maps and order statistics are coordinatewise nondecreasing. Hence $\Psi(D(x; b)) \leq \Psi(D^U(x; \beta))$ and $\Psi(D^L(y; \beta)) \leq \Psi(D(y; b))$ componentwise. Combining these inequalities with the strict lexicographic inequality proves the result. $\square$

The branch-and-bound procedure eliminates a candidate only when the displayed test succeeds. A box with one survivor contributes that winner to both inner and outer possible sets. Unresolved terminal boxes contribute all survivors to the outer set. Crucially, these contributions are combined by union over leaves: elimination on one box cannot remove a candidate that wins on another. A strict processed-box budget preserves all unprocessed leaves. This produces

$$\text{In} \subseteq S^{\text{pos}} \subseteq \text{Out}$$

with an explicitly recorded unresolved volume fraction. The new illustration uses outward-rounded enclosures and exact rational tail means and category comparisons of the outward-rounded binary endpoints. A floating-point sampled winner is a diagnostic witness; it is not by itself an exact-real optimality certificate.

Persistent ties and winners confined to boundaries can prevent finite resolution. An inner set based only on sole-winner boxes can omit such winners even when they belong to the possible set. No general finite-budget exact recovery is claimed. If every terminal box is resolved to the same sole winner, necessary optimality is certified. A positive exact-refinement margin at one nominal point alone does not establish this result.

The illustration uses the first six candidates of the saved concave instance with three criteria and replicate zero. At each $p \in \{0.005, 0.02, 0.05\}$, budgets are 64, 256 and 1,024 processed boxes. We independently sampled 256 bound vectors per level and checked that their point winners belong to every reported outer set. Outer sets were also checked to be nested as the budget increased. These are consistency checks on one instance, not a proof of soundness or a broad scalability study. At $p = 0.005$ the root box is resolved. At $p = 0.05$, more than 99% of the volume remains unresolved after 1,024 boxes. Reporting this limitation is essential to interpreting the bracket.

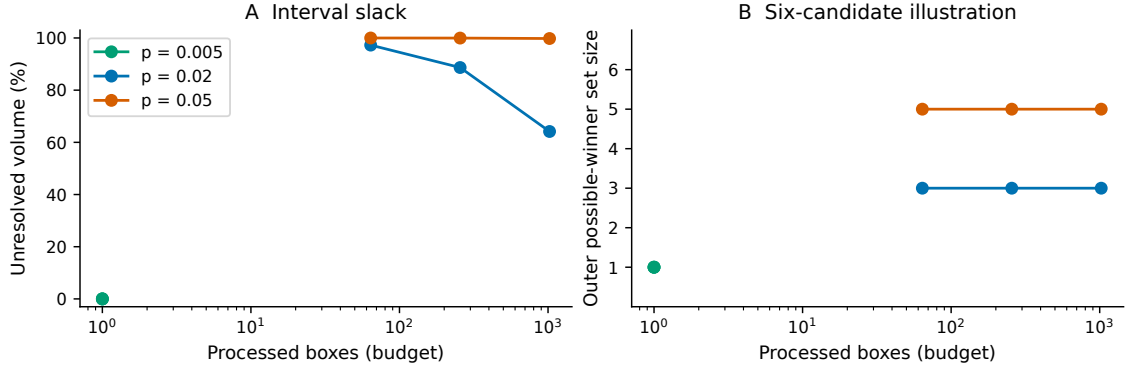


Figure 4: New interval illustration for six candidates and three criteria, with ReLexTail resolution 0.02. Each curve uses the same criterion matrix and range-relative uncertainty box. The horizontal axis reports processed boxes, which may be fewer than the budget after early termination. The narrowest box resolves immediately; larger boxes can retain substantial unresolved volume despite increased effort.

4.3 A direct formulation on a continuous polyhedron

Let $\mathcal{X}$ be a nonempty bounded continuous polyhedron, with affine criteria and nondegenerate canonical probes. Compute criterion bounds and probe anchors over $\mathcal{X}$ and then hold them fixed. Singleton and mean disappointments are affine. Maximum-probe disappointment is convex piecewise linear. Its epigraph has a linear representation. Since every subsequent objective and bound is monotone in disappointments, these epigraphs can represent the required upper bounds without introducing artificial improvements.

For a risk coordinate R_ℓ , use an integer variable $z_\ell \geq 0$ and constraint $R_\ell(x) \leq \delta_\ell z_\ell$. Minimising z_ℓ gives the least feasible category. Fix its optimum and continue to the next category. At an optimal category, a feasible point cannot have a lower actual category, because that would contradict the previous minimisation. This is why upper category bounds suffice despite the open lower cell endpoints.

For T_α , introduce $t_\alpha \in \mathbb{R}$ and $u_{\alpha q} \geq 0$, with

$$u_{\alpha q} \geq D_q(x) - t_\alpha, \quad t_\alpha + \frac{1}{\alpha K} \sum_q u_{\alpha q} \leq \delta_\alpha z_\alpha. \quad (9)$$

The maximum uses an epigraph $v \geq D_q(x)$ for every q . After the four category stages, minimise $T_{25}, T_{50}, T_{100}, M$ in sequence, retaining all earlier optima. For the terminal refinement, minimise the sums of the largest p disappointments, $p = 2, \dots, K$:

$$S_p(D) = \min_{t_p \in \mathbb{R}, u_{pq} \geq 0} \left\{ p t_p + \sum_{q=1}^K u_{pq} : u_{pq} \geq D_q(x) - t_p \quad \forall q \right\}. \quad (10)$$

Once preceding ordered coordinates are fixed, minimising S_p minimises the next coordinate. These formulations follow standard ordered-optimisation identities (Ogryczak & Śliwiński, 2003; Ogryczak & Tamir, 2003).

Proposition 3 (Sequential formulation). *For the stated continuous-polyhedral setting with $K = m+2$ nondegenerate canonical probes, ReLexTail is characterised by a finite sequence of LPs and four integer category stages. With no reuse of anchor computations and no redundancy elimination, it uses $6m + 8$ LP calls and four MILP calls.*

Proof. Criterion bounds require $2m$ LPs. The $m+1$ affine probes require $2(m+1)$ anchor LPs. Minimising the grand maximum requires one epigraph LP and maximising it requires m affine maximisations. Preprocessing therefore uses $5m+3$ LPs. Four exact risk refinements and $K-1 = m+1$ ordered-sum refinements add $m+5$ LPs. The four category stages introduce bounded integer variables. Sequentially fixing optimal categories leaves closed risk sublevel sets, and the

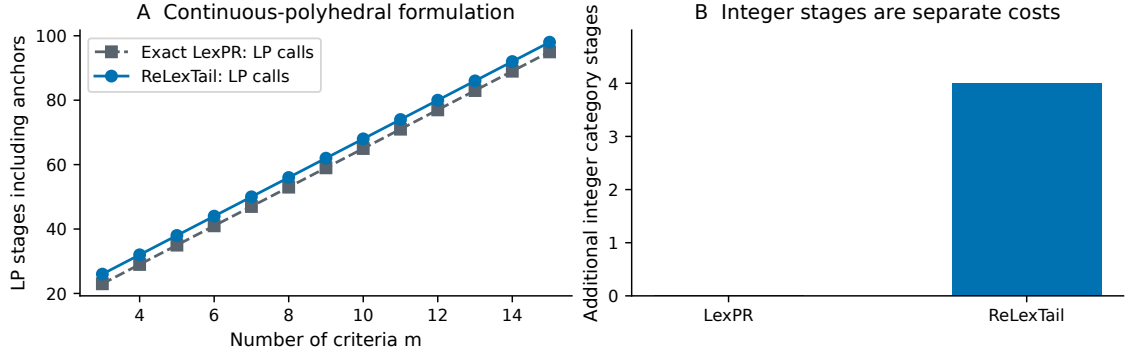


Figure 5: Theoretical solver-stage counts under the declared no-reuse continuous-polyhedral formulation. ReLexTail uses $6m + 8$ LP calls and four integer category stages; exact LexPR uses $6m + 5$ LP calls. These are formulation counts, not measured runtimes or evidence that MILP cost is constant.

later continuous objectives attain their minima on compact feasible sets. The sequence therefore realises the specified lexicographic optimum. $\square$

If the original feasible set includes integer variables, preprocessing and refinements can themselves require MILPs. If probes are dropped or redundant stages are removed, the count changes. Solver tolerances also change the operational optimisation problem and must be reported separately from resolution widths. Finally, anchors computed over a polyhedron generally differ from anchors over a finite sampled shortlist. Agreement between the resulting recommendations is not guaranteed. The present submission does not claim an independently validated end-to-end continuous MILP implementation.

5 Decision-focused certification

Section 4 certifies with a condition that asks more than the decision needs. Theorem 5 requires *every* candidate to keep *every* risk coordinate inside its cell, and Proposition 2 is applied through enclosures in which the shared normalisation anchors are widened independently for each candidate. This section replaces both with conditions matched to the question actually asked – does the recommendation survive? – proves that neither replacement weakens soundness, and measures what the change buys. The declared rule is untouched: on fixed inputs the engine of this section and the engine of Section 4 select the same alternative. What differs is what can be proved about that alternative and at what cost.

5.1 The decision region and its radius

Fix A , a retained multiset $\mathcal{Q}$, a resolution vector δ and the grid convention of Section 2. For a nominal winner x_0 that is unique, the *decision region* is

$$\mathcal{R}_{x_0} = \{b : \Psi(x_0; b) <_{\text{lex}} \Psi(y; b) \text{ for every } y \neq x_0\}, \quad (11)$$

and for the nested boxes $\mathcal{B}(p)$ of Equation (8) the *decision radius* is $\rho^*(x_0) = \sup\{p : \mathcal{B}(p) \subseteq \mathcal{R}_{x_0}\}$. The computational object is a certified inner region for $\mathcal{R}_{x_0}$, reported together with the leaves that were not resolved. Three conventions matter and are not negotiable in what follows. A qualifying set that turns out to be empty is reported as “no unique-winner certificate”, never converted into positive stability. A tied nominal optimum requires a separately defined set-preservation contract and is recorded and skipped rather than resolved by tie-breaking. And the procedure is one-sided: failure to certify at level p means only that the budget did not suffice, so an upper bound on ρ^* can come only from a verified counterexample, never from a failed search.

5.2 Certifying the decision, not every coordinate

Proposition 4 (Decision-focused soundness). *Let β be a box on which the retained multiset is fixed, and let $D^L(\cdot; \beta) \leq D(\cdot; b) \leq D^U(\cdot; \beta)$ coordinatewise for every $b \in \beta$. If*

$$\Psi(D^U(x_0; \beta)) <_{\text{lex}} \Psi(D^L(y; \beta)) \quad \text{for every } y \neq x_0,$$

then $\beta \subseteq \mathcal{R}_{x_0}$. No candidate is required to retain any category, and no coordinate of Ψ beyond the first differing one is examined.

Proof. Every coordinate of Ψ is a nondecreasing functional of D coordinatewise: order statistics, the maximum, the upper-tail means of Equation (4) and the category map B_δ of Equation (5) all are. Hence for $b \in \beta$, $\Psi(D(x_0; b)) \leq \Psi(D^U(x_0; \beta))$ and $\Psi(D^L(y; \beta)) \leq \Psi(D(y; b))$ componentwise, and componentwise inequality implies $\leq_{\text{lex}}$. The lexicographic order is total and transitive, so $\Psi(x_0; b) \leq_{\text{lex}} \Psi(D^U(x_0; \beta)) <_{\text{lex}} \Psi(D^L(y; \beta)) \leq_{\text{lex}} \Psi(y; b)$ for every $y \neq x_0$ and every $b \in \beta$. $\square$

The chain uses no retention of categories anywhere. Category retention is therefore sufficient but not necessary, and the gap is not a technicality.

Proposition 5 (The global condition can be vacuous while the decision is certain). *There exist a candidate set with rational entries, a resolution vector and a level $p > 0$ such that every candidate changes at least one categorical coordinate between the lower and the upper enclosure over $\mathcal{B}(p)$ – so μ_{cell} of Theorem 5 certifies nothing – while the test of Proposition 4 certifies the unique winner against every rival on the whole of $\mathcal{B}(p)$ without any subdivision.*

Proof. By exhibition. Example E-A of the reproducibility archive is a 5×4 matrix whose entries are integer multiples of $1/64$, hence exact in binary double precision. At $p = 0.01$ all 5 candidates leave at least one cell, so the all-candidate condition refuses the root box; the pairwise test of Proposition 4 succeeds against all four rivals at that same root box. Both facts are recomputed by `certrelex.examples.check_boundary`. $\square$

The mechanism is the one anticipated in Section 3: a coordinate may be free to move because it belongs to an alternative that is not in contention, because it sits on a boundary that is structurally invariant, or because it occurs after a comparison that is already decided. None of those motions can change the recommendation, and a condition that forbids them measures the wrong thing.

5.3 Preserving the shared normalisation anchors

Write $v_x(b) = q(r(x; b))$ for a retained aggregate probe. All candidates share the same two anchors $q^*(b)$ and $q^w(b)$ at a given b , and Equation (2) reads $D_q(x; b) = g(v_x(b), q^*(b), q^w(b))$ with

$$g(v, a, c) = \frac{v - a}{c - a}. \quad (12)$$

Widening $v - a$ and $c - a$ into two independent intervals and dividing them lets the numerator take an extreme value against a denominator that the same a could not have produced. Keeping a inside a single expression removes that.

Proposition 6 (Shared-anchor enclosure). *Let $v \in [v^L, v^U]$, $a \in [a^L, a^U]$, $c \in [c^L, c^U]$ with $c^L > a^U$. Then g is strictly increasing in v and, for each fixed v , monotone in a and in c separately, with*

$$\frac{\partial g}{\partial c} = -\frac{v - a}{(c - a)^2}, \quad \frac{\partial g}{\partial a} = \frac{v - c}{(c - a)^2},$$

each of constant sign in the variable being varied. Consequently the range of g over the product box is attained at the four corners of $[a^L, a^U] \times [c^L, c^U]$, evaluated at $v = v^L$ for the minimum and $v = v^U$ for the maximum. The resulting interval is contained in the independent quotient $[v^L - a^U, v^U - a^L] / [c^L - a^U, c^U - a^L]$, and the containment is strict whenever $a^L < a^U$ and $v^L \geq a^U$.

Proof. The two derivatives are as displayed. For fixed v the sign of $\partial g/\partial c$ is that of $-(v-a)$, which does not depend on c ; the sign of $\partial g/\partial a$ is that of $v-c$, which does not depend on a . A function that is monotone in each variable separately on a rectangle attains its extrema at the rectangle's corners: minimise first over c at an endpoint, then over a at an endpoint. Monotonicity in v places the minimum at v^L and the maximum at v^U . For containment, note that each corner value is one of the four quotients $(v-a)/(c-a)$ with $a \in \{a^L, a^U\}$ and $c \in \{c^L, c^U\}$, and every such quotient lies in the independent interval because $v-a$ lies in $[v^L - a^U, v^U - a^L]$ and $c-a$ in $[c^L - a^U, c^U - a^L]$. If $v^L \geq a^U$ the corner minimum is $(v^L - a^U)/(c^U - a^U)$, whereas the independent lower endpoint is $(v^L - a^U)/(c^U - a^L)$; the denominators differ by $a^U - a^L > 0$ with a nonnegative numerator, so the inequality is strict, and symmetrically at the upper end. $\square$

Two further identities are used and are not claimed as new. Singleton disappointments are bound-invariant by Equation (3), so m of the at most $m+2$ canonical coordinates carry no width however large the box is; the verifier of Section 5.4 rechecks that invariance at the box corners rather than assuming it. And branching is directed at the criteria whose normalised enclosure is widest among the candidates still involved in an unresolved comparison, because those are the only bound coordinates that can still move the decision; this changes where effort goes and not what is provable.

Proposition 7 (Cancellation-induced slack is not asymptotic). *There exist a rational candidate set and a level $p > 0$ at which the enclosure of Proposition 6 certifies the unique winner on the whole of $\mathcal{B}(p)$ in a single box, while the independent quotient – identical in every other respect – leaves part of the box uncertified after a budget three orders of magnitude larger.*

Proof. By exhibition. Example E-B is a 4×3 matrix over the denominator 64. At $p = 0.05$ the joint enclosure resolves the root box in 1 box; the independent enclosure has certified only 33.0% of the volume after 2,048 processed boxes. The two runs differ only in Equation (12). Recomputed by `certrelex.examples.check_slack`. $\square$

5.4 Anytime brackets, refusal and independent replay

Certified leaves are never revisited and unresolved leaves always retain their volume, so an interrupted run returns a valid certified region together with a valid bound on the uncertified volume, and continuing can only move volume from the second into the first. A box is *refused* outright when a normalising denominator or a probe active range is not certifiably positive on it; refusal leaves the box unresolved and never becomes positive stability.

Because the search is one-sided, an upper bound on ρ^* must come from elsewhere. We obtain it from an exact rational oracle that evaluates the declared rule at a single bound vector with no floating-point step: a b at which the oracle names a different unique winner is a verified counterexample and bounds ρ^* from above. Example E-C is a 4×3 rational instance whose nominal winner is candidate 1; inside $\mathcal{B}(0.05)$ there is a bound vector at which the oracle names candidate 2 uniquely, while the engine certifies $\rho^* \geq 0.0220$. The pair $[0.0220, 0.05]$ is a genuine two-sided bracket, which is what an honest anytime certificate should return in place of a single number.

Every certificate is replayed by a verifier written to be simpler than the search and independent of it: it imports nothing from the interval or enclosure modules, works entirely in exact rational arithmetic, and knows nothing about gates, budgets or branching. It is handed a leaf and a claimed winner and recomputes the enclosure and the comparison. A leaf both routes accept has been certified twice, by different arithmetic. The full numerical contract – reference rule, operational selector, certification arithmetic and verifier, and what each does *not* claim – is recorded in the archive.

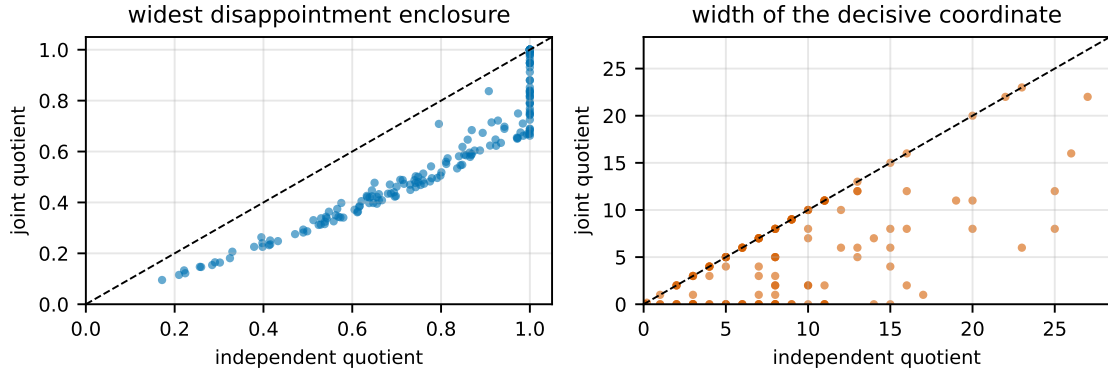
5.5 Fixed-budget certification experiments

The study is a fixed-rule comparison: the same declared rule, probes, uncertainty box, arithmetic contract, box budget and hardware in every arm, so that any difference is attributable to the certification machinery and not to a different recommendation. Instances come from six development families and two locked stress families (near-tied margins; correlated heavy-tailed measurement

Table 3: Ablation ladder. Every arm certifies the same rule on the same inputs with the same budget of 300 processed boxes.

Arm	Sufficient condition	Quotient	Branching
A0	all-candidate retention	independent	widest coordinate
A1	decision-focused	independent	widest coordinate
A2	decision-focused	shared anchors	widest coordinate
A3	decision-focused	shared anchors	decision-relevant

Dependency preservation tightens the comparison that decides

**Figure 6:** Dependency preservation tightens the comparison that decides. Each point is one instance at the root box $\mathcal{B}(0.02)$; the dashed line is equality. Left: widest disappointment enclosure. Right: enclosure width of the profile coordinate at which the nominal winner and its closest rival first differ. Points below the diagonal are instances where keeping the shared anchors inside one quotient narrows the enclosure.

noise) at $m \in \{3, 6, 10\}$ with eight replicates each, 192 candidate sets in all, certified on a shortlist of 10 alternatives. The split is a deterministic function of the replicate index and was fixed in source before any run; the 90 instances of the earlier sections remain the development baseline and are not used as confirmatory evidence. Resolution is $\delta = 0.02$ throughout and the focal level is $p = 0.02$. Each arm adds exactly one component to the one above:

The primary endpoint is the certified lower bound ρ^{lo} on the decision radius, obtained by bisection at the matched budget; it is continuous, paired within instance, and one value per arm. The two prespecified contrasts are $P1 = A1 - A0$, isolating the decision focus, and $P2 = A2 - A1$, isolating dependency preservation, both on the locked test split with Holm correction inside that family. Every other contrast is labelled exploratory. The independent unit is the instance. Because the study was not preregistered with a third party, p -values are descriptive.

Correctness (E0). Across 192 instances, 11,072 bound vectors were drawn from inside certified leaves and evaluated by the exact rational oracle, and 5,029 certified leaves were replayed by the independent verifier. There were 0 falsified certificates and 0 replay failures. Testing supplements Propositions 4 and 6; it does not substitute for them, and a zero count is evidence of the absence of the errors searched for, not of soundness in general.

Mechanism (E1). At the root box the widest disappointment enclosure averages 0.774 under the independent quotient and 0.582 under the shared-anchor quotient. The coordinate that decides the nominal comparison is categorical on 176 of the 178 instances with a resolvable root box; on those, the median enclosure width of that coordinate falls from 5.0 categories to 0.5. The remaining instances are decided on a tail mean and are excluded from that statistic because the two are not in the same units. The share of instances certified at the root box with no subdivision at all rises from 34.8% to 52.8%. Figure 6 shows the paired comparison.

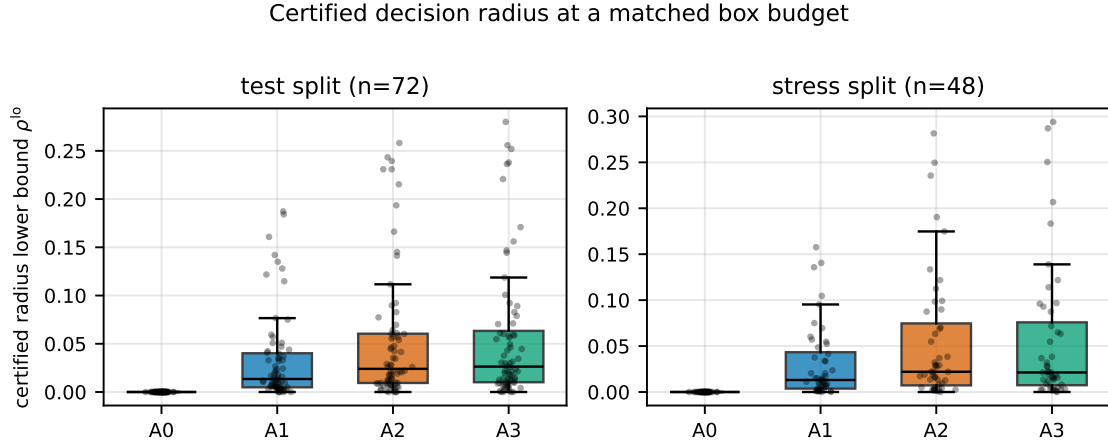


Figure 7: Certified lower bound on the decision radius at a matched budget of 300 processed boxes, on the two locked splits. A0 is the all-candidate retention condition, A1 adds decision focus, A2 adds shared-anchor enclosures, A3 adds decision-relevant branching. Boxes show quartiles with the median; points are individual instances. A value of zero means “not certified within this budget”, not a radius of zero.

Fixed-budget certification (E2). On the locked test split (72 instances), mean ρ^{lo} is 0.0000 for A0, 0.0330 for A1, 0.0534 for A2 and 0.0556 for A3. The share of instances with a strictly positive certified radius moves from 0.0% to 94.4%. The prespecified contrasts are

$$P1 = 0.0330 [0.0233, 0.0437], \quad P2 = 0.0204 [0.0150, 0.0264],$$

with 95% paired bootstrap intervals over instances, win/loss/tie counts 67/0/5 and 68/0/4, probabilities of superiority 0.965 and 0.972, and Holm-adjusted p -values $1.52\text{e-}12$ and $1.52\text{e-}12$. Decision-relevant branching contributes a further 0.0023 [0.0001, 0.0043], reported as exploratory. On the locked stress split the corresponding means are 0.0000 and 0.0555. Figure 7 shows the distributions.

Anytime behaviour (E7) and brackets. Mean certified volume of $\mathcal{B}(0.02)$ after 64, 256 and 1,024 processed boxes is 0.000, 0.000 and 0.000 for A0 against 0.529, 0.557 and 0.589 for A3. A verified switch witness was found for 136 of 192 instances; among those the median bracket width is 0.083 in p , and the number of instances whose certified lower bound exceeded its own witness – which would be a contradiction – is 0. Figure 8 shows both.

What this does not show. These are certification results and nothing more. They do not show that ReLexTail chooses better than another method, that its recommendations carry lower external regret, or that any calibration guarantee holds; the corresponding tracks were not run, and Section 7 records them as unestablished. The instances are generated, the shortlist size is declared rather than elicited, and the stress families test extrapolation without licensing any claim about arbitrary deployment shift. A radius reported as zero always means “not certified within this budget”.

6 Computational study and supplier illustration

6.1 A matched, reproducible protocol

The study isolates the behaviour of the finite selection rule. It comprises 90 synthetic candidate sets: three front geometries, three criterion counts ($m = 3, 6, 10$), and ten independent replicates per combination. Each set contains 80 candidates. Linear fronts use $f = 0.5s$ for a uniformly Dirichlet-distributed simplex vector s . Convex fronts use $f_i = s_i^2$. Concave fronts normalise a vector of square-root-transformed independent uniform draws to Euclidean norm one. These are controlled

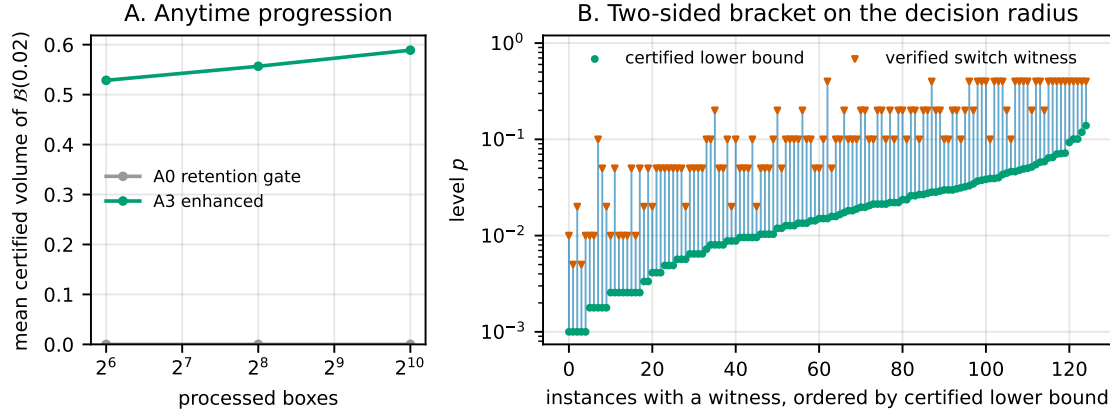


Figure 8: Anytime behaviour and the two-sided radius bracket. Panel A: mean certified volume of the focal box against the processed-box budget, for the all-candidate retention condition and for the enhanced engine; certified volume is nondecreasing in the budget because certified leaves are never revisited. Panel B: for each instance with a verified switch witness and a strictly positive certified radius, the interval between the certified lower bound and the smallest level at which the exact rational oracle names a different unique winner. A bar’s height is what remains unresolved between proof and counterexample, not an error bar.

synthetic geometries, not exact solutions of named engineering problems or real procurement datasets. Every generated matrix is saved.

The nominal criterion bounds and probe anchors are computed from the candidate set. The family consists only of singletons, the mean and the maximum, subject to the stated retention test. Retention is checked to remain identical on every sampled perturbation. Eight common resolution widths are tested: 0.001, 0.005, 0.01, 0.02, 0.03, 0.05, 0.075, 0.1. The comparators are exact LexPR on D ; exact leximax and Chebyshev selection on the normalised criterion vector; Mean- D ; upper-25% and upper-50% tail minimisation on D ; and a uniform weighted sum on the normalised criteria. They separate changes in order, tail aggregation and representation. Chebyshev and leximax happen to select identically throughout the saved population, so only leximax is shown in the compact table. All 15 method variants remain in the data. These are controlled comparators, not an exhaustive ranking of MCDA methods.

At each level $p \in \{0.01, 0.025, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$, 500 uniform bound perturbations are shared across all methods. The same random directions are scaled across levels. This yields 45,000 matched selections per method and level-specific summaries for every instance. Within an instance, one common set of 10,000 Dirichlet(1) weight vectors evaluates every candidate on the fixed nominal criterion scale. The master seed is 20260905, and instance streams are indexed by geometry, criterion count and replicate. Deterministic ties use the smallest row identifier.

For a test weight vector w , the linear loss and its range-normalised regret are

$$L_w(x) = \sum_i w_i r_i(x; b_0), \quad \Delta L_w(x) = \frac{L_w(x) - \min_{y \in A} L_w(y)}{\max_{y \in A} L_w(y) - \min_{y \in A} L_w(y)}. \quad (13)$$

If the denominator is zero, regret is defined as zero for all candidates for that weight. Mean regret averages all 10,000 values. Upper-tail regret averages exactly the 2,500 largest values. The latter is empirical upper-25% CVaR, including the appropriate mass when values tie. Candidate losses are computed once on the nominal scale. We then evaluate the points selected under perturbed bounds and average first over draws within each candidate set, then equally over candidate sets.

Point-change rate is the fraction of draws changing the fully refined point relative to its nominal identifier. Category-set change rate compares the complete nominal and perturbed W_{cat} sets. We also save category-set size, Jaccard overlap and retention of the nominal point. None of these set metrics is substituted for a point-change rate in cross-method comparisons. Nominal divergence compares the nominal ReLexTail and LexPR identifiers.

Table 4 and Figure 9 use the same 90-instance population at $p = 0.05$. Confidence intervals resample paired instance summaries, preserving each instance’s methods and all 500 perturbations

Table 4: Paired comparison on 90 candidate sets at $p = 0.05$. Regret is evaluated on the nominal scale after perturbed selection. The interval is a 95% paired instance-bootstrap interval for the tail-regret difference from LexPR. Point changes are percentages.

Method	Mean regret	Tail regret	Difference interval	Point changes
Weighted-sum	0.2915	0.6270	[+0.05677, +0.11376]	19.52
Mean-D	0.3070	0.5491	[+0.00009, +0.01275]	30.50
Leximax	0.4136	0.5340	[-0.01216, -0.00556]	35.64
Tail25-D	0.3800	0.5401	[-0.00473, -0.00097]	20.84
LexPR	0.3810	0.5428	[+0.00000, +0.00000]	20.88
ReLexTail $\delta = 0.005$	0.3810	0.5425	[-0.00092, +0.00011]	19.96
ReLexTail $\delta = 0.01$	0.3810	0.5419	[-0.00181, -0.00025]	20.16
ReLexTail $\delta = 0.02$	0.3799	0.5411	[-0.00324, -0.00064]	19.30
ReLexTail $\delta = 0.05$	0.3741	0.5407	[-0.00362, -0.00093]	21.21
ReLexTail $\delta = 0.075$	0.3729	0.5401	[-0.00422, -0.00129]	20.78
ReLexTail $\delta = 0.1$	0.3731	0.5407	[-0.00401, -0.00054]	23.24

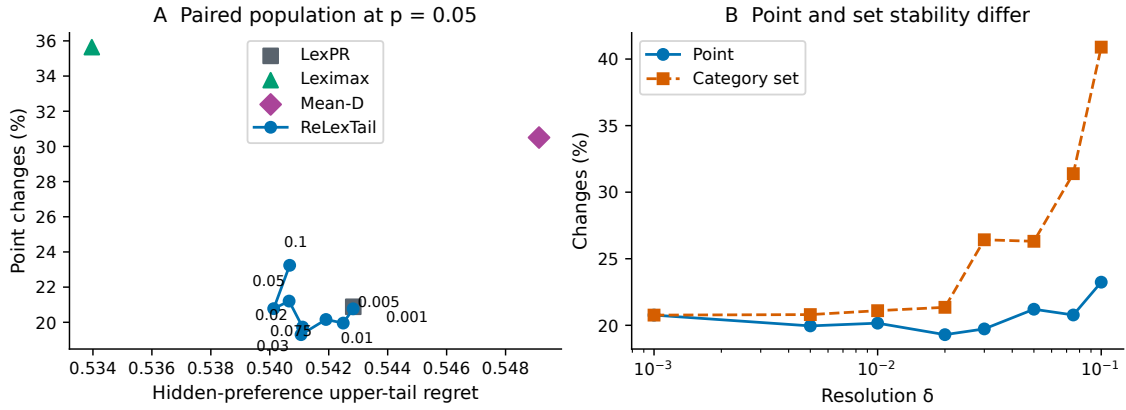


Figure 9: Matched quality-stability results at $p = 0.05$ on all 90 synthetic instances. Lower values are preferred on both axes in panel A; labels on the ReLexTail curve give resolution widths. Panel B separates changes to the point from changes to the category-optimal set. Connecting lines identify tested resolutions and do not imply monotone behaviour between them.

as one block. Table intervals use 10,000 percentile-bootstrap replicates. For two focal outcomes at $\delta = 0.02$, Table 5 uses 20,000 paired bootstrap replicates, a paired Wilcoxon signed-rank check, probability of superiority with half credit for ties, and Holm correction across exactly those two checks. Seed 20260906 fixes both resampling analyses. The focal width represents two percentage points of active range and was retained from the earlier protocol; the analysis was not preregistered, so inferential quantities remain descriptive. No resolution is declared universally optimal.

6.2 Results and limits of the trade-off

At $\delta = 0.02$, ReLexTail changes its point on 19.30% of draws compared with 20.88% for LexPR. The paired mean difference is -1.584 percentage points, but its 95% bootstrap interval $[-3.482, 0.091]$ includes zero and the Holm-adjusted Wilcoxon p -value is 0.148. Thus the enlarged run does not support a clear point-stability improvement. Mean external upper-tail regret is 0.5411 versus 0.5428; the paired difference is -0.00177 with interval $[-0.00329, -0.00065]$, superiority probability 0.722 and adjusted $p = 4.44 \times 10^{-7}$. This effect is consistent on this population but small in magnitude and conditional on the held-out weight distribution.

Exact leximax obtains lower tail regret (0.5340) but a higher point-change rate (35.64%). Tail25-D attains tail regret 0.5401 and a 20.84% point-change rate, showing that most of the numerical tail benefit does not require categories. The uniform weighted sum attains the lowest mean regret (0.2915) and a much worse upper-tail regret (0.6270), illustrating that average performance and

Table 5: Two focal paired effects for ReLexTail at $\delta = 0.02$ minus LexPR ($p = 0.05$, 90 instance summaries). Probability of superiority counts a lower ReLexTail value as a win and gives half credit to ties. W/L/T are wins/losses/ties. Holm correction covers the two displayed Wilcoxon checks.

Outcome	Mean difference	95% bootstrap interval	Superiority	W/L/T	Holm p
Point changes (pp)	-1.584	[-3.482, +0.091]	0.572	32/19/39	0.148
Upper-tail regret	-0.002	[-0.003, -0.001]	0.722	48/8/34	4.44×10^{-7}

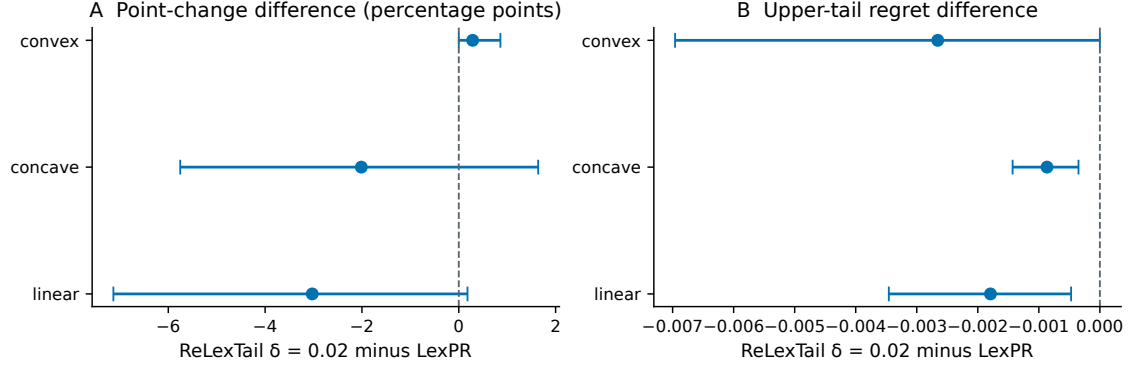


Figure 10: Family-specific paired differences between ReLexTail at $\delta = 0.02$ and LexPR, at $p = 0.05$. Each family contains 30 candidate sets spanning three criterion counts. Points are mean differences; bars are 95% percentile-bootstrap intervals from 4,000 paired instance resamples. Negative values favour ReLexTail. The intervals are descriptive and not adjusted for multiple comparisons.

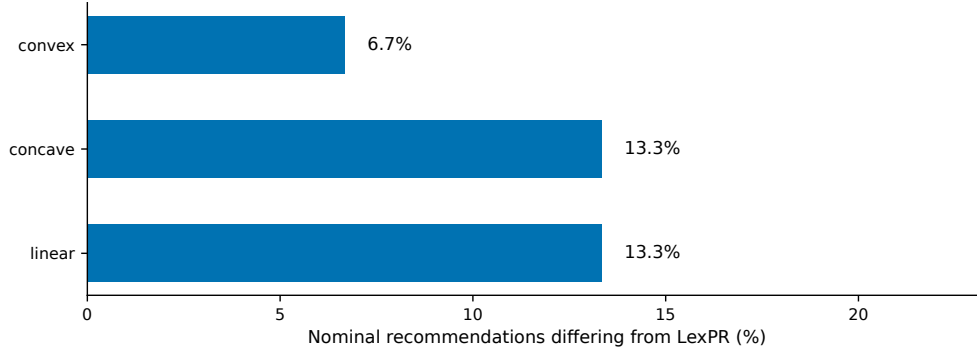


Figure 11: Nominal recommendation disagreement between ReLexTail at $\delta = 0.02$ and exact LexPR, by synthetic geometry. Each denominator is 30 candidate sets. This concerns unperturbed recommendations and is distinct from the point-change rates under perturbed bounds.

protection across hidden preferences are different objectives.

Coarser resolution is not uniformly more stable. At $\delta = 0.1$, ReLexTail’s point-change rate is 23.24%, above the LexPR rate, even though its mean tail regret is slightly lower. Category-set change rises to 40.90%, with a mean perturbed category-set size of 2.90. At $\delta = 0.02$, category-set change is 21.35% and mean size is 1.08. A larger set is therefore not automatically a more stable set under exact set equality. The category-stability theorem supplies a conditional local guarantee, not an empirical promise that these sets never change.

Figure 10 disaggregates the main comparison rather than assuming the pooled effect holds for every geometry. Figure 11 separately reports nominal disagreement. A method can share a nominal winner with another method yet respond differently to perturbed bounds. Conversely, disagreement alone does not establish better quality. These distinctions prevent nominal selection, external regret and robustness from being treated as interchangeable outcomes.

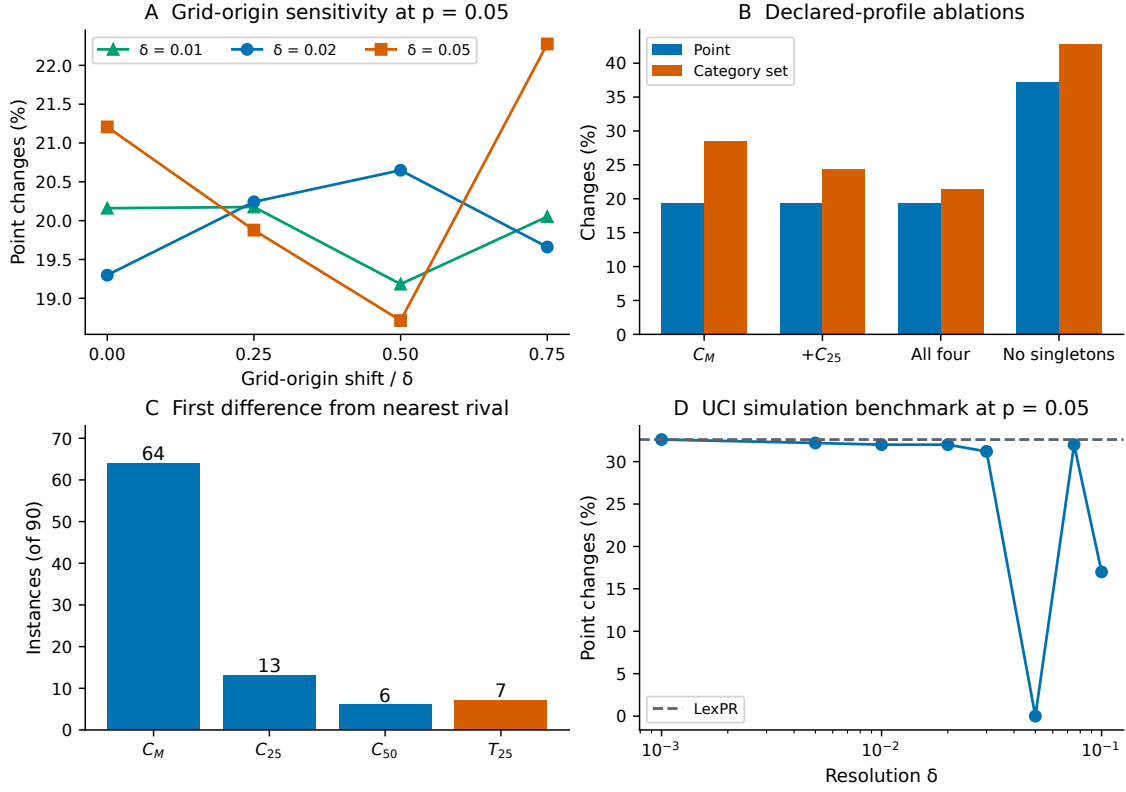


Figure 12: Additional checks using the frozen synthetic directions and a public simulation dataset. Panel A shifts fixed-cell origins at three resolutions. Panel B changes the category prefix or removes singleton probes. Panel C identifies the first full-profile difference from the nearest nominal rival. Panel D is a single UCI building-simulation instance and must not be read as population evidence; its nonmonotone jump illustrates why widths require sensitivity analysis.

6.3 Grid, representation and public-data checks

Figure 12 tests three potential alternative explanations. Across four grid origins, point changes for $\delta = 0.02$ range from 19.30% to 20.65%; the result is therefore sensitive in magnitude but not created by the unshifted origin. Using only C_M , then C_M, C_{25} , or all four category coordinates gives similar point-change rates (19.40%, 19.40% and 19.30%) while category-set changes fall from 28.54% to 21.35%. Removing all singleton probes changes the representation and raises point changes to 37.22%; it is not a neutral simplification. Against each nominal winner’s nearest full-profile rival, the first difference occurs at C_M on 64 instances, C_{25} on 13, C_{50} on six and exact T_{25} on seven. No later coordinate is decisive in this population.

For an application-oriented reproducibility check, we downloaded the UCI Energy Efficiency dataset (Tsanas & Xifara, 2012). It contains 768 simulated building designs, not measurements from occupied buildings. We minimise surface area, heating load and cooling load; surface is only a material-use proxy, not a monetised or environmental impact. Removing duplicate objective profiles and Pareto-dominated rows leaves four unique candidates. On the unperturbed front, LexPR, Mean-D, the uniform weighted sum and every tested ReLexTail width select saved source row 26. At $p = 0.05$, LexPR changes on 32.6% of 500 draws and ReLexTail at $\delta = 0.02$ on 32.0%; at $\delta = 0.05$ ReLexTail happens not to change, whereas neighbouring widths do. This isolated discontinuity demonstrates boundary dependence, not superior real-world performance. The archived source, licence, checksum, filtering and all perturbed results are included.

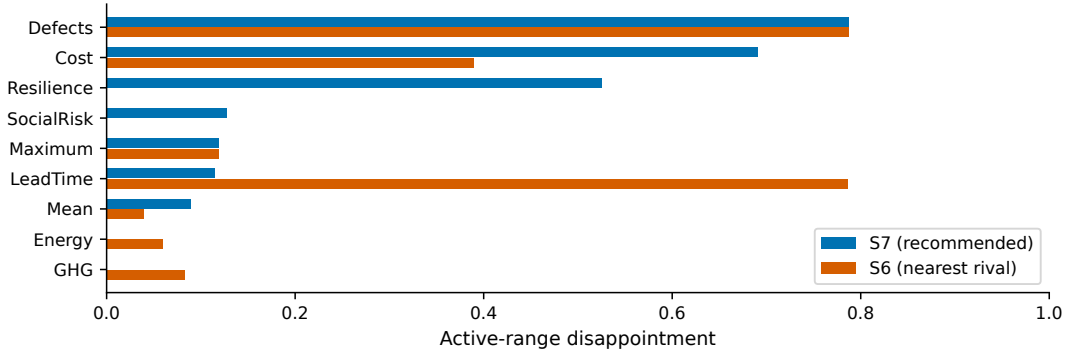


Figure 13: Labelled probe disappointments for the recommended supplier S7 and its nearest rival S6. Values are recomputed from the saved criterion matrix with nine canonical probes. Bars explain the individual concerns; optimality follows from comparing complete profiles across all ten candidates.

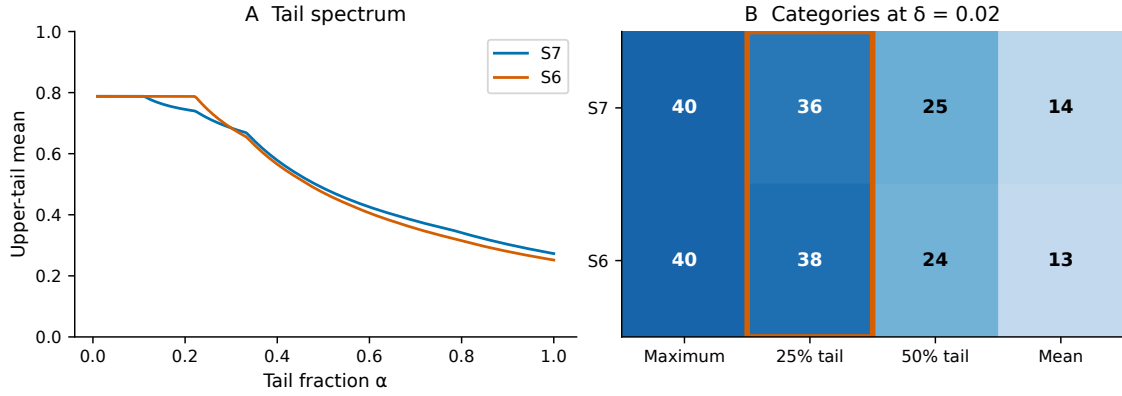


Figure 14: Supplier tail spectrum and category comparison at $\delta = 0.02$. S7 and S6 tie in maximum category, but S7 has a lower upper-25% tail category, highlighted in panel B. Later coordinates need not favour S7 once that first strict difference settles the comparison.

6.4 A constructed supplier decision

The supplier illustration contains ten named alternatives and seven criteria to minimise: unit cost, defect rate, lead time, greenhouse-gas emissions, energy intensity, social risk and supply-resilience risk. Values are fixed in the accompanying source and are constructed, not field measurements. The instance illustrates how a decision analyst can expose the decisive concern and investigate sensitivity. It does not validate the model’s preferences for a procurement organisation.

With nine retained canonical probes and common resolution $\delta = 0.02$, ReLexTail selects S7. Figure 13 compares its labelled disappointments with S6, the nearest rival under the full profile. Both have maximum disappointment 0.787879, hence maximum category 40. Their upper-25% tail means are 0.715814 and 0.743146 respectively, giving categories 36 and 38. Thus the decisive coordinate against S6 is C_{25} , not C_M . S1 has maximum category 41 and is eliminated earlier. Reporting only the S7–S1 contrast would miss the closest rival and misrepresent the actual hierarchy.

The distance $0.800000 - 0.787879 = 0.012121$ is only S7’s upward distance to its maximum-category boundary. It is not the global μ_{cell} of the stability theorem. On this instance the global minimum distance is zero because some risk coordinates lie exactly on boundaries. The theorem’s sufficient condition is therefore uninformative without a more refined treatment of structurally invariant coordinates. A positive bound-robustness guarantee must come from a direct certificate rather than that single distance.

At $p = 0.05$, three of the 300 draws change the nominal supplier. At $p = 0.20$, the rate is 30.67%. S1 and S7 are the only sampled winners from $p = 0.05$ through $p = 0.50$. This does not prove that no other supplier can win in those boxes. Sampling and the interval illustration in Section 4 use

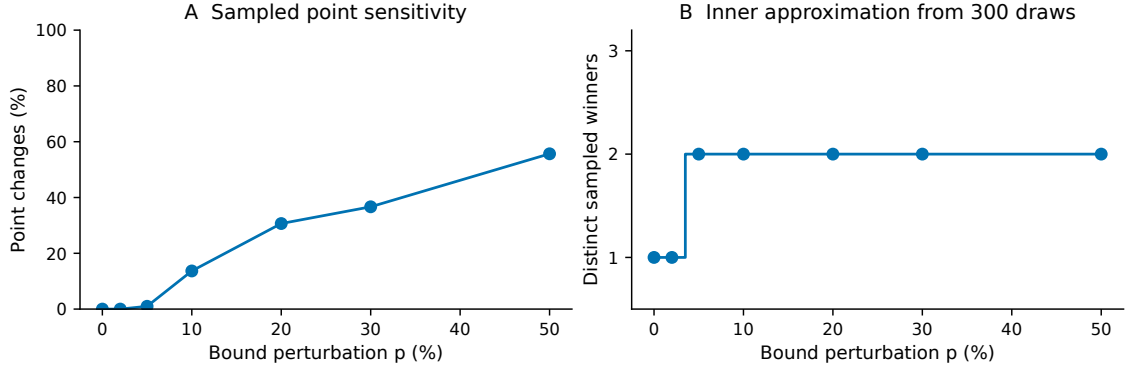


Figure 15: Supplier sampling sensitivity for ReLexTail at $\delta = 0.02$, using 300 independent uniform bound draws at each level and nominal-range perturbations. Panel A reports point changes. Panel B reports distinct observed winners, an inner sampling approximation rather than a certified outer bound. The $p = 0.5$ endpoint is sampled only and is excluded from closed-box certification.

the same type of range-relative perturbation, but they concern different candidate matrices and must not be combined into a single empirical certificate.

7 Discussion and conclusions

ReLexTail offers a specific form of noncompensatory decision aiding: a deterioration to a worse maximum category cannot be offset by improvements elsewhere, while differences inside that category may be deferred in favour of broader-tail concerns. This is a preference commitment. An analyst should elicit whether that commitment is appropriate before selecting the probe family or widths. The method is most useful when named concerns, finite resolution and a reproducible explanation are important parts of the decision process.

Resolution is not a statistical confidence level or a solver tolerance. It sets category widths on the active-range disappointment scale. A width of 0.02 means two percentage points of a probe’s available range, not two percentage points of a physical criterion. Equal widths across risk coordinates are convenient for comparison but need not fit every application. A practical report should record the elicitation exercise, state the chosen widths, compare nearby widths and explain whether category membership or the final point is the intended recommendation. The four-origin analysis shows that the focal point-change rate remains within a narrow two-percentage-point band, but other widths and the UCI example remain visibly boundary-dependent. Grid sensitivity is therefore a required diagnostic, not a proof of robustness.

The finite-set properties are conditional on the declared representation. Positional anonymity does not make the rule preference-free, since repeated probes change their influence. Pareto compatibility requires retained separating probes. Active anchors introduce dependence on the candidate set. Category stability concerns perturbations that avoid every relevant boundary and does not imply stability of exact refinements. The method’s auditability comes from exposing these conditions and replaying its comparisons, not from eliminating modelling choices.

The computational study shows modest tail-regret gains over exact LexPR at some widths and deterioration in point stability at a coarser width. The enlarged paired analysis does not establish a point-stability gain at $\delta = 0.02$, because its interval includes zero. Tail25-D also captures much of the external tail-regret difference, so the categories should be justified by resolution semantics and set-valued reporting rather than by that metric alone. These findings argue for reporting the full trade-off rather than selecting a favourable metric. The three synthetic geometries and uniform linear test preferences restrict external validity. The public building data are themselves simulated and reduce to four nondominated objective profiles under our chosen objectives; they provide reproducibility and a boundary stress test, not stakeholder or field validation. Operational applications, additional utility families and prospective elicitation studies would be needed to establish practical benefit.

Table 6: Claim-to-evidence register. “Proved” means a proof appears in this paper; “measured” means the number is produced by the released scripts from released data; “unestablished” means the paper does not claim it.

Claim	Status	Evidence
ReLexTail is a complete transitive preorder with exact refinement	proved	Section 3
Conditional Pareto compatibility and replication invariance	proved, conditional	Section 3, stated assumptions
Category stability under a sup-norm perturbation	proved, sufficient only	Theorem 5
Interval elimination is sound	proved	Proposition 2
A decision can be certified with no category retained	proved	Proposition 4
The all-candidate condition can certify nothing while the decision is certain	proved by exhibition	Proposition 5, rational example
Shared-anchor enclosures are contained in independent ones	proved	Proposition 6
That containment can decide certification	proved by exhibition	Proposition 7, rational example
No emitted certificate was falsified	measured	Section 5.5, exact oracle and independent replay
Decision focus and dependency preservation raise the certified radius	measured, locked split	Section 5.5, paired contrasts P1 and P2
Upper-tail regret differs from exact LexPR at $\delta = 0.02$	measured, small	Section 6, paired interval
Point-stability gain at $\delta = 0.02$	not supported	paired interval includes zero
Sequential polyhedral characterisation	proved	Section 4
End-to-end continuous solver	unestablished	no validated implementation
Lower external regret at matched coverage	unestablished	track not run
Calibrated risk or coverage guarantee	unestablished	track not run
Benefit on measured operational data	unestablished	no field data with repeated observations
Priority over the closest literature	unestablished	full-text verification pending

Interval analysis addresses a different uncertainty question. Its sound outer set can remain large even when random draws repeatedly select one point. This is an informative distinction between absence of observed switches and proof that switching is impossible. Section 5 narrows that gap along the two axes where the earlier engine was conservative for reasons unrelated to the decision. Requiring all candidates to retain all categories is sufficient but not necessary, and Proposition 5 exhibits an instance where it certifies nothing while the recommendation is certain over the whole box; widening the shared normalisation anchors independently is sound but loose, and Proposition 7 exhibits an instance where that slack alone decides whether the box is certified. Neither change touches the declared rule, and neither is a new result in interval analysis: what is new is the identification of which shared quantities occur in this rule and the measured consequence for certification. The continuous-polyhedral formulation establishes a computational characterisation, while the independent numerical validation of a full solver implementation remains future work.

The scope of the certification evidence should not be overstated. It concerns generated instances on a declared shortlist size, with a resolution and a focal level fixed in advance; a certified radius reported as zero means only that the budget did not suffice. Nothing in it bears on decision quality. Whether the recommendation carries lower external regret than a competing method at matched coverage, and whether a calibrated emission policy can meet a prespecified risk target, are separate statistical claims that require a separate experimental track with independent decision episodes; that track was not run here, and no part of the certification result substitutes for it. The upper-tail regret comparison reported in Section 6 remains a small, metric-dependent effect on one synthetic population. Table 6 states, for each claim the paper makes, what kind of evidence supports it.

The paper establishes a complete and transitive resolution-aware preorder, conditional Pareto

compatibility, replication invariance, a local category-stability guarantee, and a decision-focused dependency-preserving certificate with a two-sided radius bracket and an independent replay path. Its numerical evidence identifies a qualified, metric-dependent quality trade-off on the stated synthetic population rather than a general stability advantage, together with a clearer certification result on a separate locked benchmark. The supplier illustration makes the decisive upper-tail comparison explicit. Together these results support ReLexTail as a transparent modelling option when the resolution of declared concerns matters, with sensitivity, unresolved uncertainty and the untested claims reported alongside the recommendation.

Declarations

Author contributions (CRediT). Madani Bezoui: conceptualization, methodology, software, formal analysis, investigation, visualization, writing—original draft, and writing—review and editing.

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Competing interests. The author declares no competing interests.

Ethics. No human participants, animals or personal data were involved. The supplier data are constructed.

Data and code availability. The submission source package includes the scripts, generated candidate matrices, perturbation directions, protocol, per-instance results, statistical summaries, figure sources and the UCI Energy Efficiency file under CC BY 4.0 with its SHA-256 checksum. The UCI dataset is independently available at <https://doi.org/10.24432/C51307>. The earlier public project archive is available at <https://doi.org/10.5281/zenodo.21771786>, and the project repository is <https://github.com/MadBezoui/ReLexTail>. The revised experiment and figures are identified by the accompanying submission package and must not be attributed to an unchanged earlier release. Publishing a new immutable DOI for this exact revision remains an author-side release action. The complete local reproduction command is documented in the package README.

Use of artificial intelligence. An AI assistant assisted with manuscript revision, code preparation, numerical consistency checks and figure preparation. Reported numerical results were computed by the supplied deterministic and seeded scripts. The author is responsible for checking and approving the scientific content and all submission declarations.

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Figure legends

- 1 Overview of the decision procedure and graphical abstract. The declared model determines active-range disappointments. Category comparisons precede exact refinement. Under uncertain bounds, sampling and interval certification answer different questions and require different reports. 2
- 2 Fixed resolution cells and boundary sensitivity, illustrated at $\delta = 0.05$. Filled endpoints belong to the lower category and open endpoints to the upper category. Scores 0.049 and 0.051 are close but occupy different cells. This is a constructed explanation of the rule, not an empirical result. 5
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- 11 Nominal recommendation disagreement between ReLexTail at $\delta = 0.02$ and exact LexPR, by synthetic geometry. Each denominator is 30 candidate sets. This concerns unperturbed recommendations and is distinct from the point-change rates under perturbed bounds. 18

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